

Making Health Data ACTIONABLE



Shepherd 

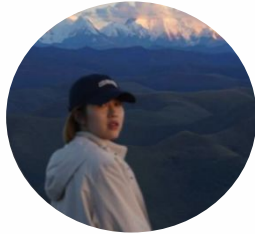
An Always-on context-aware autonomous AI assistant



Meet our members



Mridul Kumar Sharma



Zhangxizi Qiu



Haoming Liu



Wing Chun Leung

Why Shepherd 🐑?



Designed to guide and protect users -- like a shepherd guiding and protecting a flock, the system watches over users:

- filtering misinformation and
- providing timely, actionable guidance in their daily lives

RECAP



Motivations #1

Post-COVID misinformation overload

BOOM Research

Fake News In The Time Of Coronavirus: A BOOM Study

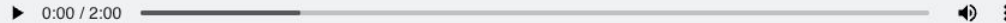
We analysed 178 fact checks on COVID-19 related misinformation from January to May. This is what we found.

By - Archis Chowdhury |

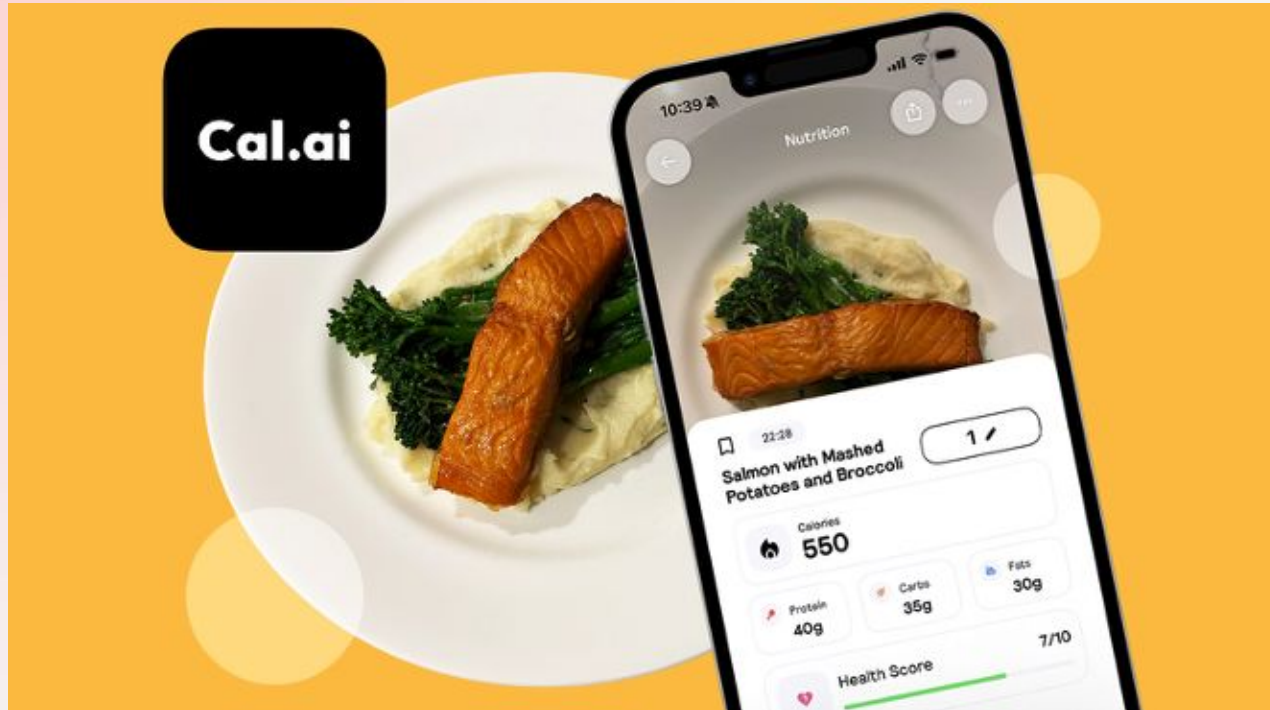
Published - 8 May 2020 10:33 AM • 2 mins read



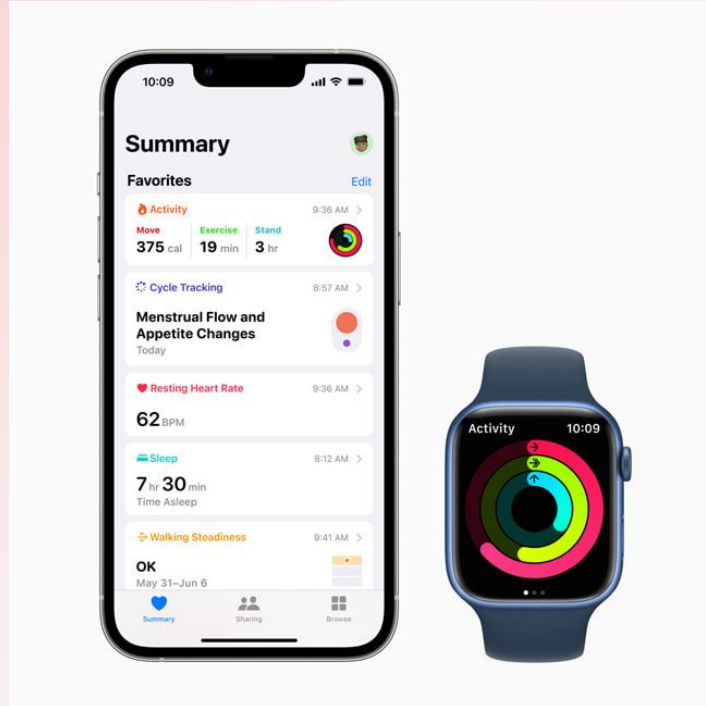
Listen to this Article



Motivations #2 - Post-COVID passive health tracking, no actionable insights



Motivations #2 - Post-COVID passive health tracking, no actionable insights



So What?





Motivations

01

Misinformation overload

User Needs:

- Trustworthy and Reliable
- Timely Intervention

02

No actionable insights

User Needs:

- Actionable Guidance
- Personalization

In additional

- Minimize Cognitive Load
- Remain minimally intrusive (Calm Interaction)



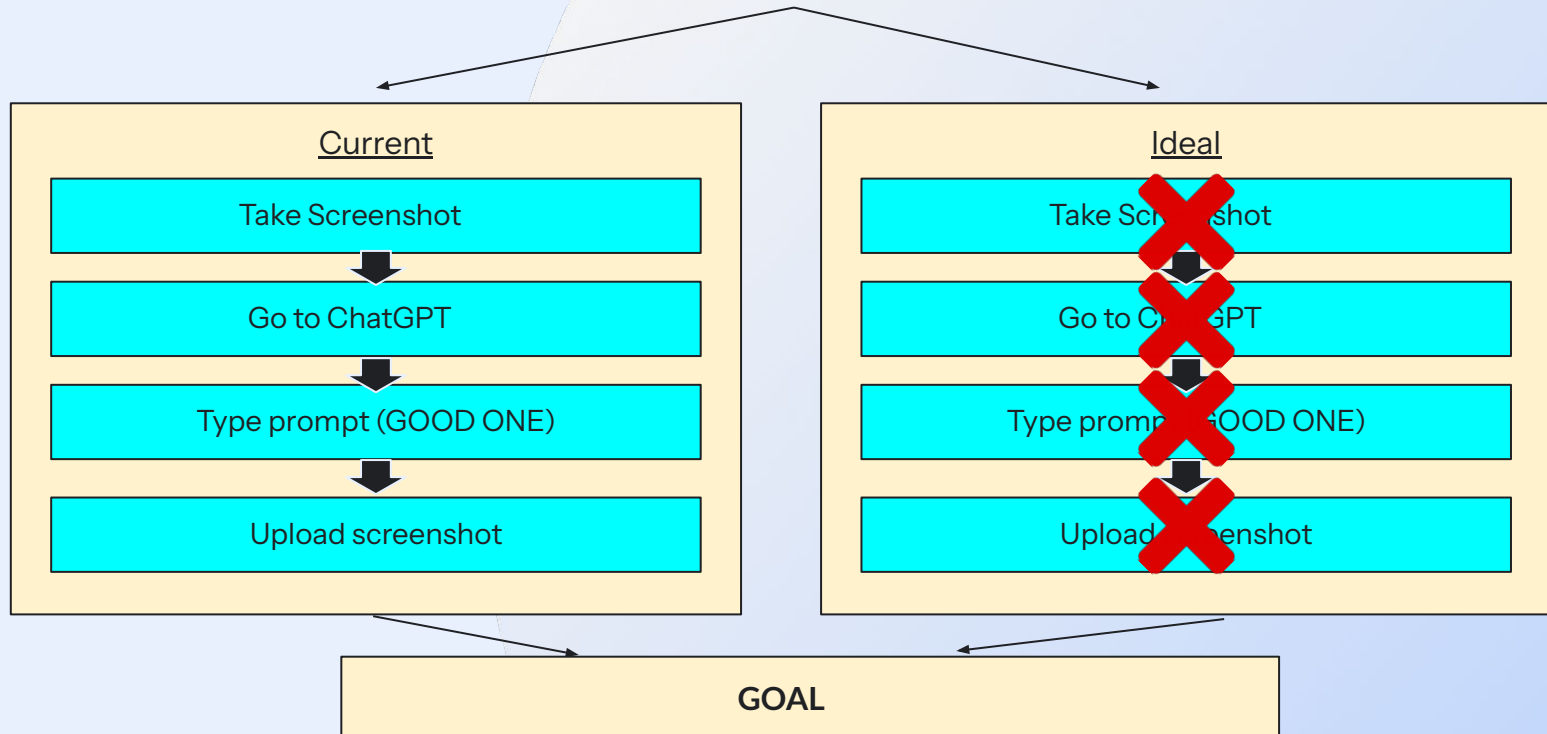
BRAINSTORMING



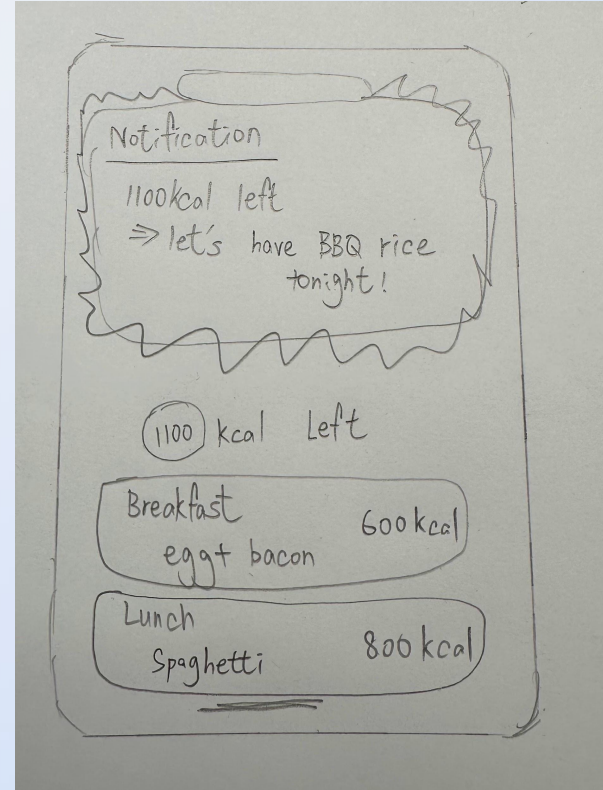
GOMS

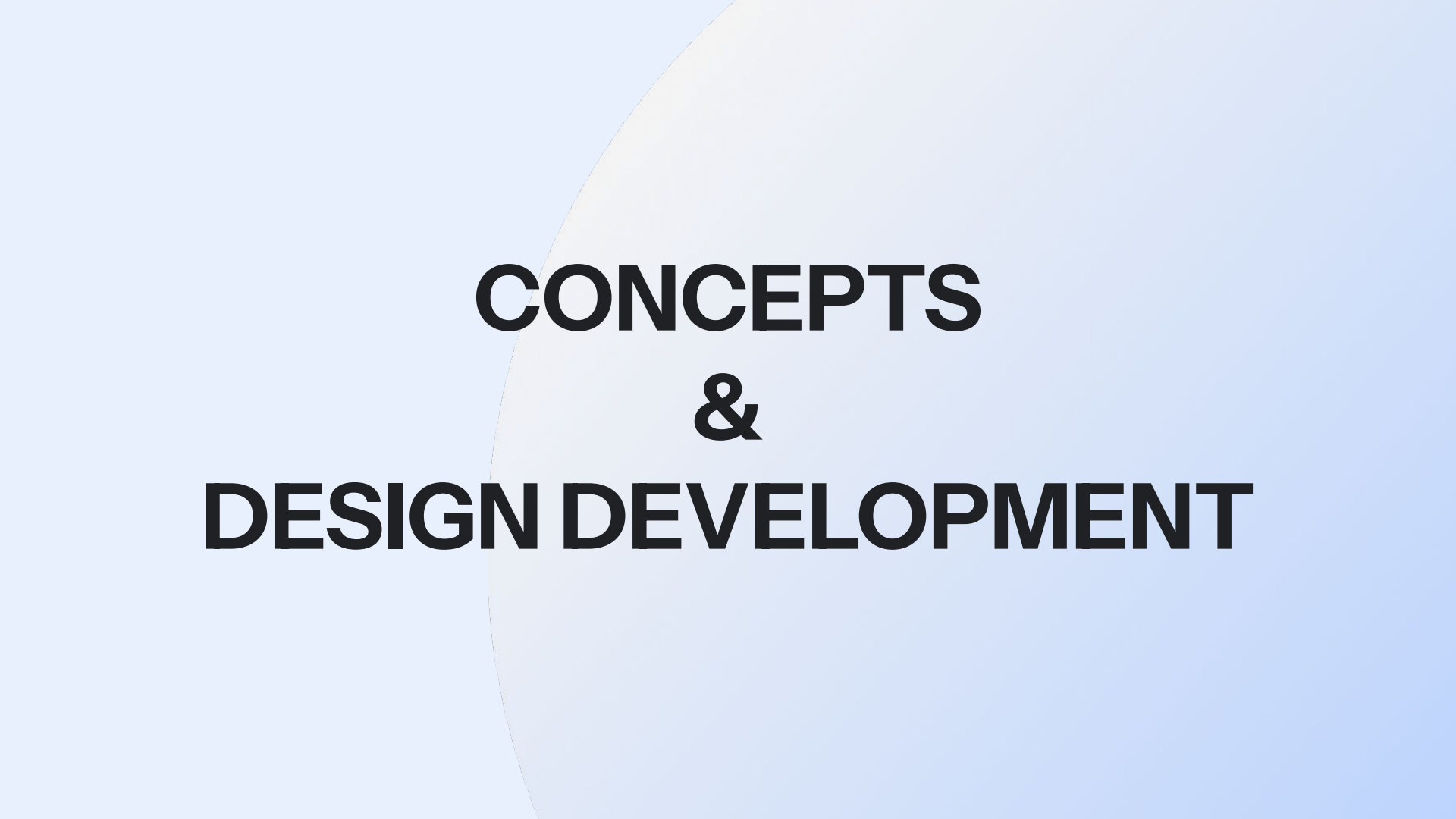
(Goals, Operators, Methods, and Selection rules)

Raw data (fake news, health data, etc.)



Low-Medium Fidelity





CONCEPTS & DESIGN DEVELOPMENT

Ideation and Concept Selection



Diverse Concept Exploration

- **Brainstormed** multiple interaction ideas, scanning, manual entry, and voice logging.
- Compared approaches for speed, user effort, and clarity.



Image-Based Analysis

- AI-powered image recognition, minimizing manual input.
- **Delivers explainable results, boosting user confidence and trust.**



Selection Criteria

- Prioritized **usefulness, feasibility**, and **novelty** to align with user needs.

Design choice

Manual Food Lookup Frustrations

- Users spend time searching nutrition tables and guessing calorie amounts.

Effortless AI-Powered Scanning

- Simply upload a food image for instant analysis and clear calorie breakdown.

Boosts Confidence and Speed

- Reduces effort and delivers **fast, actionable results** for informed meal decisions.



Design and Iteration Journey



Rapid Prototyping Evolution

- Started with low-fidelity sketches to map user flows
- Progressed to digital wireframes for refined layouts



Iterative User-Centered Testing

- Feedback-driven improvements at each stage
- Peer walkthroughs validated interaction quality



Functional Prototype Features

- Implemented scan animation for visible AI processing
- Macro breakdown display supports quick decision-making

System Flow

Seamless Food Image Upload

- Users initiate calorie analysis by uploading a food photo.

AI-Powered Visual Analysis

- It processes the image to estimate calories and macros with **minimal user effort**.

Transparent System Feedback

- Clear interface cues and scan animation keep users informed of system progress.

Actionable Results Delivery

- Concise breakdown empowers quick, confident food decisions.

Key Features Overview



Instant Image Upload & Preview

- Users visually confirm their food selection, reducing errors and boosting **confidence**.



Visible AI Scan Animation

- Processing is made transparent, keeping users informed and enhancing **trust** in the system.



Clear Calorie & Macro Breakdown

- Nutritional data is presented in easy-to-read cards for **quick, actionable decisions**.

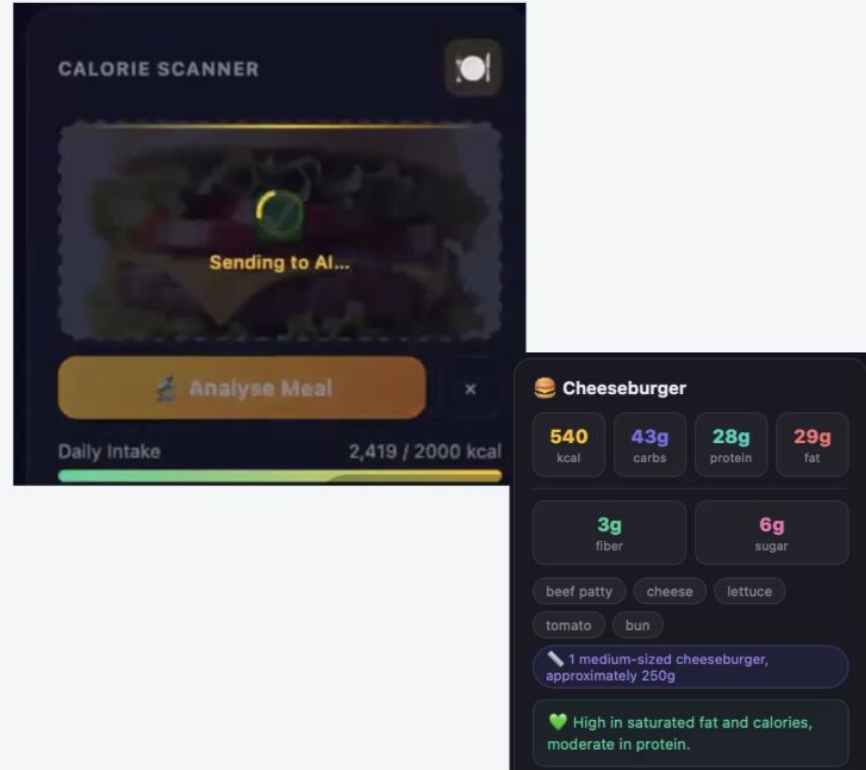


Guided Results with Progressive Disclosure

- Relevant information shown first, with deeper details available as needed to avoid overload.

Innovative Design Decisions

- Upload preview -----> reduces mistakes
- Scan animation -----> builds trust
- Macro cards -----> simplify data interpretation
- Concise layout -----> avoids overload



Balancing AI and User Trust



Streamlined User Experience

- Minimal steps from upload to result ensure ease of use.



AI as Support, Not Replacement

- System guides decisions while users maintain **control**.



Transparent Processing

- Visible scan animation and clear explanations build **trust**.



STEP COUNT



8,262

of 10,000 steps

83% to goal

5.2 km

Distance

342

Cals Burned

1h 12m

Active Time

98 bpm

Avg HR

Daily progress



SLEEP MONITOR

7h 23m

11:32 PM - 6:55 AM

87

Score

Deep · 1h 52m

Light · 2h 48m

REM · 1h 45m

Awake · 58m

LIVE NOTIFICATIONS

14:16

Tuesday, May 5

14:16

NEW



AI Lens - FakeGuard

just now

⚠ Misinformation Detected

"New vaccine causes 5G activation"

94% Fake



SmartLens - NutriAI

12m ago

Lunch Analysed 🍽

Chicken Caesar Salad · 487 kcal

+487 kcal



SmartLens - FitTrack

1h ago

8,000 Steps Reached! 🏃

Only 2,000 more to hit your daily g...

80% Complete



SmartLens - SleepAI

7h ago

Sleep Report Ready 📊

7h 23m · Score 87/100 · Good

Excellent Rest

EARLIER



AI Lens - FakeGuard

3h ago

✅ News Verified

FAKE NEWS DETECTION

AI Lens · Scanning live

"New vaccine causes 5G activation in bloodstream"

Fake



94%

socialmedia.io · 2m ago

"WHO confirms new seasonal flu strain"

Real



97%

who.int · 3h ago

"Scientists find water traces in Mars atmosphere"

Unsure



61%

newsblog.net · 1h ago

1 fake today

12 verified

CALORIE SCANNER





Interview (brief, 5 Questions)

Likert-scale questions (1-5)

- Does the system provides useful insights?
- Does the recommendations are actionable?

Feature-specific questions (Yes/ No)

- I would rely on this system to judge post credibility ?
- I would follow the suggested actions ?

Open-ended

- What would you improve?



x5



User feedback

Too many
notifications...



Good user

1. Usefulness / Value



2. Actionability



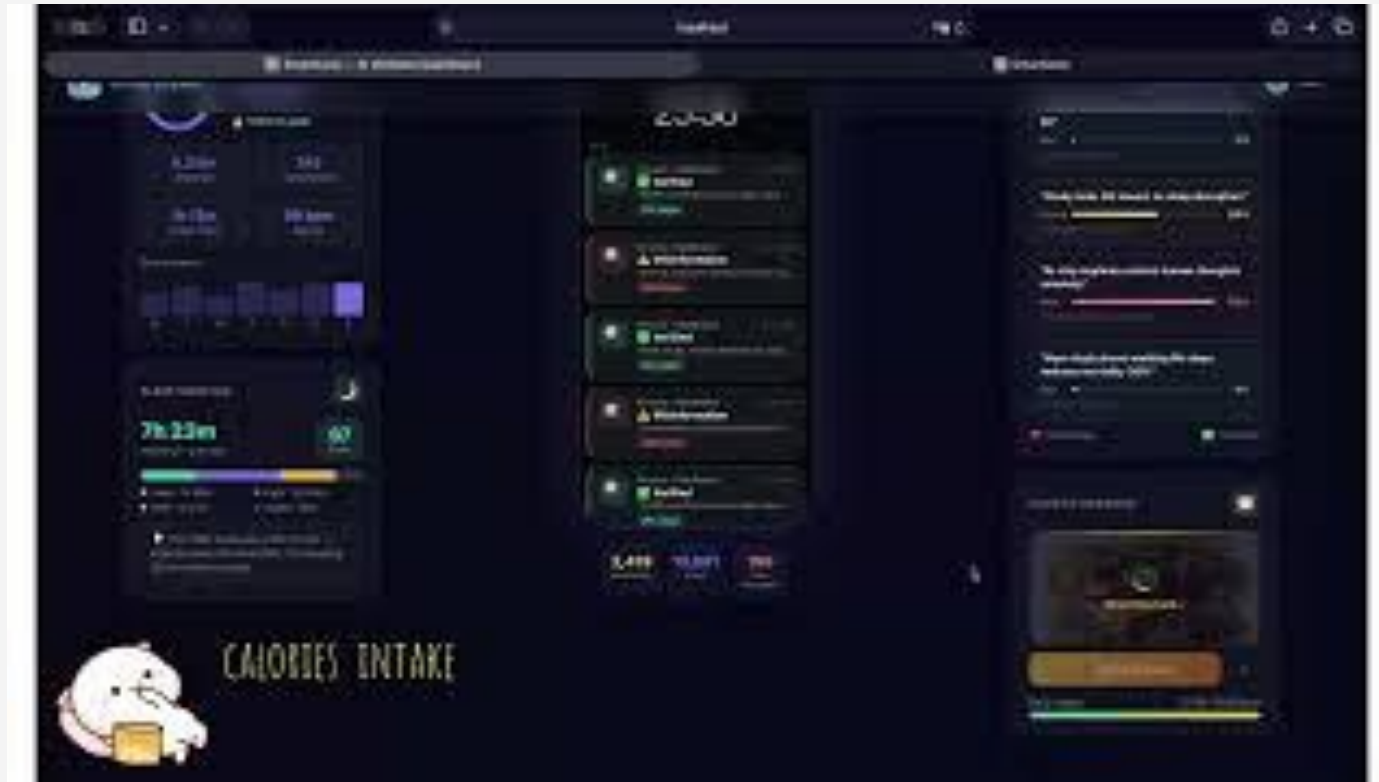
3. Cognitive Load (Ease of Understanding)



4. Trust & Accuracy



DEMO





FEATURES & DESIGN DECISIONS

Highlighting Our Innovative Features

While many health apps exist, we've focused on creating truly innovative features that address core HCI challenges. These are the key differentiators that make Health Agent a game-changer.

Fake-News
Automatic
Detector

Actionable &
Personalized
Health Guidance

Context-Aware
Inference

FEATURES

CORE DIFFERENTIATORS

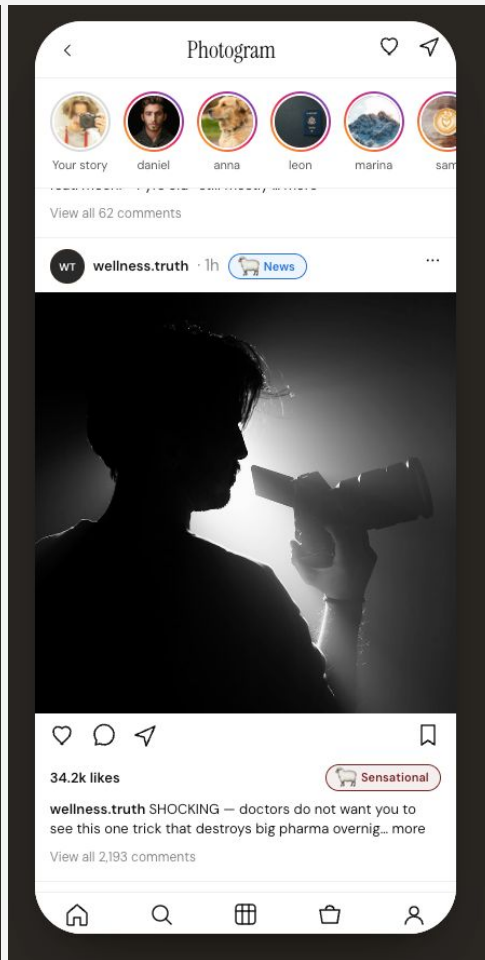
Beyond the basics, we've built unique features that set Health Agent apart. Every element is designed to make health management simpler, smarter, and more human.

Innovative Features

01. Fake News Automatic Detector

User Needs: Navigating the overwhelming amount of health information on social media is challenging. Users need a way to quickly identify credible sources and avoid misinformation.

Our Innovation: Health Agent integrates with social platforms like Instagram and Threads. It automatically categorizes posts into 'social' or 'news' and labels them as 'credible', 'sensational', or 'mixed', helping users make informed decisions.

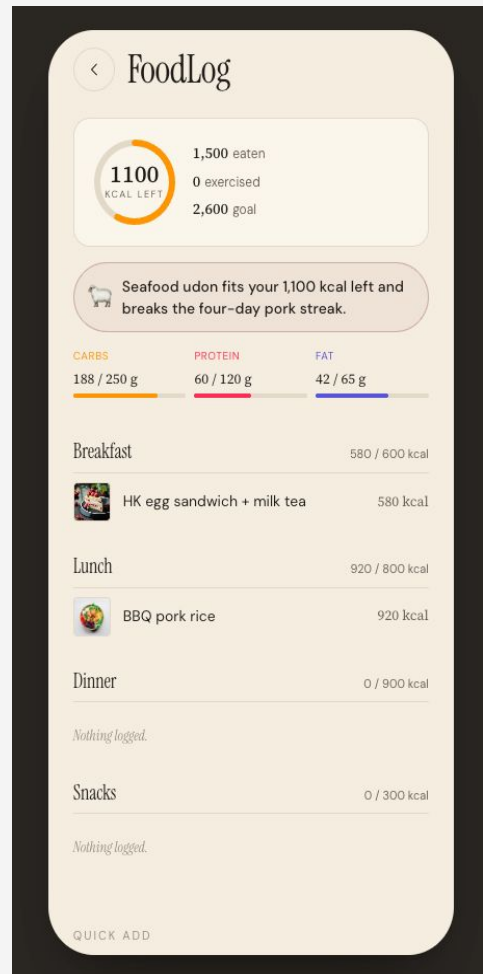
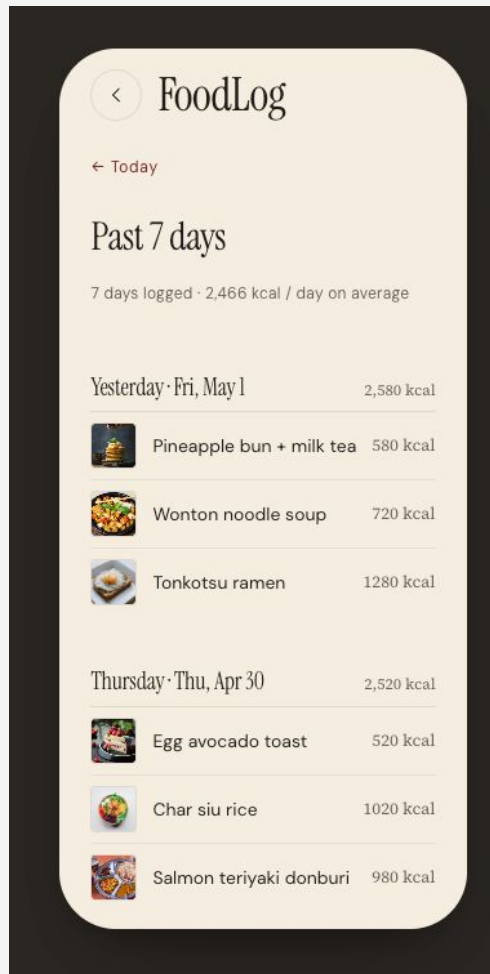


Innovative Features

02. From Passive Tracking to Actionable and Personalized Health Guidance

User Needs: Health trackers provide data, but users often don't know how to act on it. They need personalized, actionable advice to improve their habits.

Our Innovation: Health Agent analyzes your calorie and dietary data over time. It doesn't just show numbers; it provides specific, personalized suggestions for your next meal, turning passive tracking into an active health improvement journey.

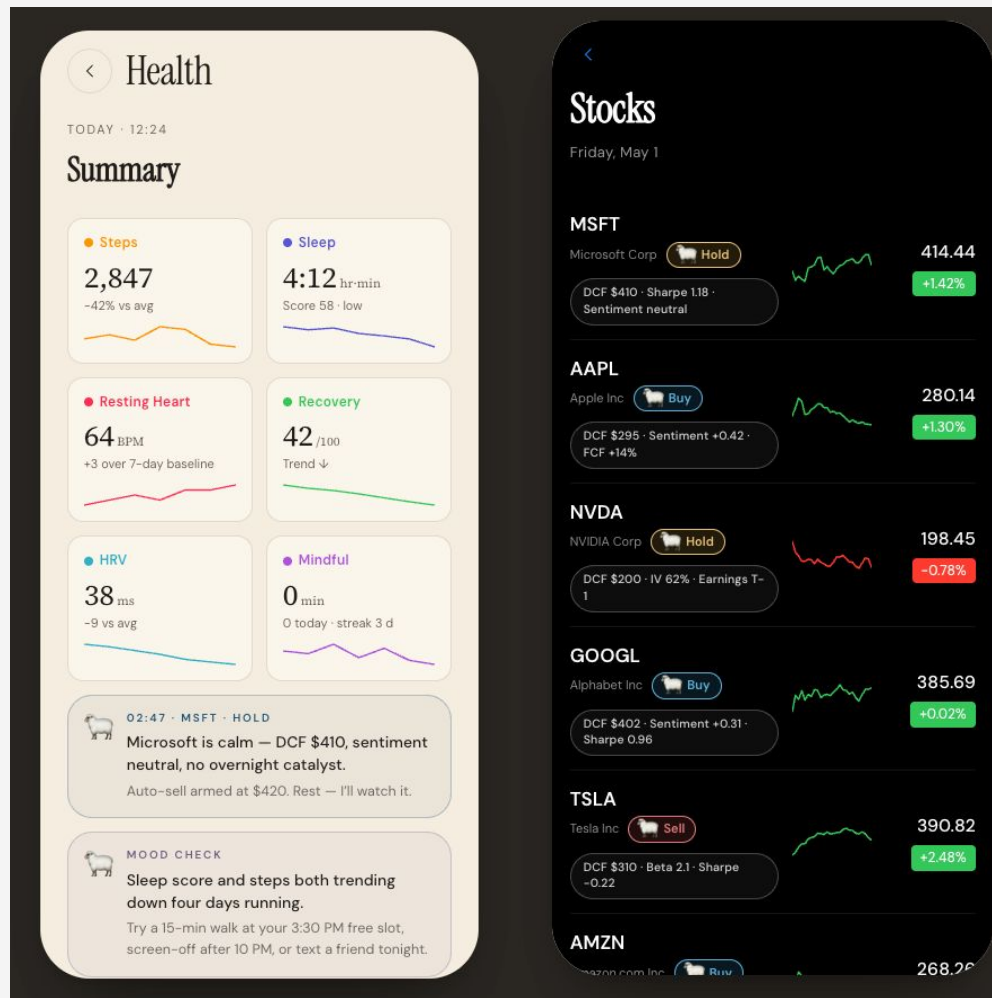


Innovative Features

03. Context-Aware Inference

User Needs: Health issues are often linked to non-health factors like stress. Users need an agent that can understand these hidden connections.

Our Innovation: Health Agent connects the dots between your health data and daily activities. For example, if it detects poor sleep and inactivity, and notices you're checking stocks at night, it infers financial anxiety and can suggest resources to help manage stress.



Justifying Key Design Decisions

1. User-Centered Design: From "Need" to "Solution"

Our entire design process was rooted in **User-Centered Design**. We began with rigorous needfinding, identifying two critical user pain points:

- Post-COVID “infodemic” (overwhelm of misinformation)
- The lack of actionable guidance from health data

Every feature, from the Fake News Detector to non-intrusive interaction, was designed to directly address these needs, ensuring the product is desirable and useful.

CORE PRINCIPLES

Desirability > Technological Feasibility

User Needs > Business Wants

Clarity > Complexity

Justifying Key Design Decisions

2. Privacy-by-Design: Building Trust from the Core

Rationale:

Health data is inherently sensitive. We prioritized embedding Privacy-by-Design principles at the product's core, rather than as an afterthought.

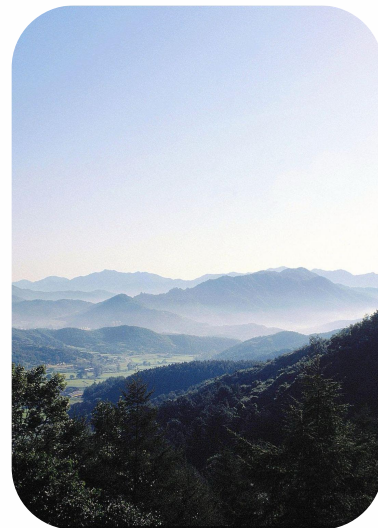
Implementation & Impact:

The agent leverages a secure long-term memory system with strict data protection protocols. We also provide explicit, one-click privacy controls to empower users with full ownership of their data. This approach fulfills ethical HCI requirements and is the foundation for establishing enduring trust.



Core Principles

- Embed Privacy-by-Design from day one
- Secure long-term memory system & protocols
- One-click privacy controls for user autonomy
- Align with ethical HCI standards



[Privacy Policy](#)

Justifying Key Design Decisions

3. Quiet & Trustworthy Interaction: Non-Intrusive Design

User Needs: Users suffer from "alert fatigue" and information overload. They need an assistant that respects their focus and doesn't constantly interrupt.

Implementation: We designed Health Agent to be non-intrusive, operating quietly in the background, analyzing data and providing support without constant notifications.

CORE PRINCIPLES

- Non-intrusive interaction
 - Respect user focus
- Minimize cognitive load

EVALUATION & DISCUSSION

Evaluation - Questionnaire Design



We interviewed five friends across different age groups. Our application imposes **no age limitations** and targets all health-conscious individuals. The age of our respondents ranged from 18 to 75 years old.

Evaluation - Questionnaire Design



* Photos are not of the actual interviewees

We interviewed 5 friends and family members across age groups. Our application imposes **no age limitations** and targets all health-conscious individuals. The age of our respondents ranged from 18 to 75 years old.

Evaluation - Questionnaire Design

Questionnaire Chart

Part 1	Brand Introduction & Questionnaire Instructions
Part 2	Q1 Cross-Device Data Integration / Q2 Truth Engine / Q3 Quiet Operation
Part 3	Q4 Multimodal Interaction / Q5 Privacy and Data Security Trust
Part 4	Q6 Open-ended Feedback (Text Input + Quick Tags)
Part 5	Closing: Thank You Page + Privacy Statement

Evaluation – Questionnaire Feedback



For Q1 and Q2, the majority of participants selected ratings of 4 or 5, which demonstrates **high overall satisfaction** with these two features.

For Q3, young and middle-aged participants were generally satisfied, while elderly users expressed dissatisfaction because they are more health-sensitive and prefer to stay updated on their health conditions in real time.

For Q4, all participants selected the context-adaptive option, indicating a strong preference for **environment-aware alerts**.

For Q5, younger individuals exhibit lower privacy sensitivity, whereas privacy requirements rise steadily as age increases.

Q6: Open Feedback - Additional Features & Suggestions



Personalized meal plans based on health data (e.g, protein, carbs, fiber).



Mental health detection feature



Mental health reminder feature



Medications reminder features



Upload and analyze past health reports

5

Generate exercise recommendations

Discussion

Key Insights: The Value of Shepherd AI Agent



Post-Pandemic Health Focus:
attention to health



Feasibility Question:
Demand for a
realize product



Broadenes:
Health across
age groups (groups)



Agent Swacends:
Health focus
across all age
(elderly & young)

4

Agent Expectation:
Desire for AI assistance
in meal plans & healthy
lifestyles

5

Conclusion:
Shepherd AI Agent
is highly meaningful
and valuable

Key Learnings from the Course



Scientific & User-Centered Development the
ability to develop AI
products scientificall
and user-centered



UI/UX Principles in AI Era:
Human-centered
design principles,an
applicable despite
AI tools



UI/UX Design in AI Era:
Human-centered
principles remain
applicable despite
changing AI tools



Product Design & Quality:
Learned how to design a
product and what makes a
good product.



Final user feedback

1. Usefulness / Value



A large, light blue circle is positioned on the right side of the image, partially overlapping the white background. The circle's edge is smooth and its color is a soft, pale blue.

THANK YOU